



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. ~~202602010101~~.....

Name of the Program and semester: B.Tech and IV Sem

Name of the Course : Deep Learning

Course Code: TCS-464

Time: 3 hours

Maximum Marks: 100

Note:

- All the questions are compulsory.
- Answer any two sub questions from a, b and c in each main question.
- Total marks for each question is 20 (twenty).
- Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

a. Define the following terms in the context of neural networks:

- Perceptron
- Activation function
- Loss function

(CO1)

b. Consider a simple Multi-Layer Perceptron (MLP) with one hidden layer.

Explain the process of **forward propagation** step-by-step and analyze how activation functions influence the output of the network. Provide a small numerical or schematic example. (CO3)

c. Compare different **optimization algorithms** (Gradient Descent, Stochastic Gradient Descent, Adam) used in training neural networks.

Which optimizer would you choose for training a deep MLP and why?.

(CO4)

Q2.

(2X10=20 Marks)

a. Explain the basic architecture of a CNN with a neat diagram description. (CO2)

b. Consider an image classification task using a CNN.

Explain how the **convolution and pooling layers** work step-by-step to extract features from an input image.

Analyze how increasing the number of filters affects image recognition performance. (CO3)

c. Compare CNNs with traditional fully connected neural networks for image processing tasks.

Evaluate why CNNs perform better in image recognition applications.

Design a simple CNN architecture for classifying handwritten digits (like MNIST), specifying layers such as convolution, pooling, and fully connected layers. (CO4)



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Q3. (2X10=20 Marks)

a. Define the following terms in the context of Recurrent Neural Networks (RNNs):

- a) RNN (Recurrent Neural Network)
- b) Vanishing gradient problem
- c) LSTM (Long Short-Term Memory network).

(CO1)

b. Consider a sentence processing task using an RNN.

Explain how the **vanishing and exploding gradient problems** occur during training.

Analyze why standard RNNs struggle with long-term dependencies in sequential data. (CO3)

c. Design a simple LSTM-based architecture for a text prediction or sentiment analysis task. (CO4)

Q4. (2X10=20 Marks)

a. Explain in simple terms how an **Autoencoder** works.

Also, describe the basic idea behind GANs (**Generator and Discriminator**).

(CO1)

b. Give one real-world application each for:

- a) Autoencoders
- b) GANs
- c) Transfer Learning

(CO2)

c. What is **Transfer Learning** in deep learning?
Explain its advantage with one simple example.

(CO1)

Q5. (2X10=20 Marks)

a. What is Natural Language Processing (NLP)? Give one simple real-world application of NLP. (CO1)

b. What is an autonomous vehicle? Mention any two technologies used in self-driving cars. (CO2)

c. What is medical image analysis in deep learning? Give one example where it is used in healthcare. (CO2)